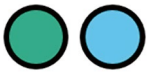
CHAPTER 14

ADM Techniques



This chapter is part of the GREEN and BLUE reading tours. See Figure 1-1 in Chapter 1, Section 1.3, for alternative reading tours.

A number of topics that were once an integral part of the TOGAF Framework have been split off and given their own place in this chapter. This includes the broader topics of Architecture Principles, Stakeholder Management, and Risk Management. But topics such as architecture patterns, gap analysis, interoperability requirements, Business Transformation Readiness Assessments, and Architecture Alternatives and Trade-Offs have also found their way into this chapter.

The reason for this is that they describe a specific application. One that is invoked, so to speak, at several points in the implementation of the TOGAF Standard.

The unbundling of these topics in turn contributes to the modularity that The Open Group has achieved with the publication of the Standard.

14.1. Technique or No Technique

The Architecture Development Method is often regarded as the core of the TOGAF Standard, providing a structured approach to developing and managing Enterprise Architectures. It consists of a cyclic method that helps organizations align business and IT needs effectively. Various techniques are applied throughout the Architecture Development

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Method cycle to ensure efficient decision-making, risk mitigation, and stakeholder engagement. These techniques support different phases of the Architecture Development Method and enhance architectural practice.

When the TOGAF Standard talks about techniques, it does not always mean a *specific way of doing things*. What is really meant by techniques is an *elaboration of a topic* or component *that is frequently used* within the framework or during the implementation of the Architecture Development Method. A recurring topic that requires further explanation and clarification in order to be properly applied.

Some topics are essential aspects of developing an architecture. However, they are not necessarily a specific technique. Take Stakeholder Management or Risk Management, for example. Both are essential and of course should not be forgotten or overlooked. But they are not techniques. Otherwise, one could also argue that Architecture Roadmapping is a technique. Or creating a Baseline Architecture. No, some topics that are listed as techniques are simply not.

Conducting a Business Transformation Readiness Assessment, on the other hand, is. So is performing a gap analysis. The reason for this is that the Business Transformation Readiness Assessment can be seen as a complement to an existing process or activity. In fact, Phases A (Architecture Vision) and E (Opportunities and Solutions) examine an organization's ability to implement change. Conducting a Business Transformation Readiness Assessment serves to complement these activities.

The same applies to the gap analysis. Phases B, C, and D (Business, Data/Application, and Technology Architecture, respectively) establish the Baseline and Target Architectures. To get from the Baseline to the desired Target Architecture, it is necessary to create (candidate) roadmap items. This activity is complemented by a gap analysis.

In short, not all of the topics that the TOGAF Standard refers to as techniques are actually techniques. This does not change the fact that all

of them are essential to the correct execution of architecture activities and the development of an architecture.

That said, the TOGAF Standard distinguishes between eight *techniques*:

- **Architecture Principles:** These are fundamental rules and guidelines that influence architecture development and governance. They ensure consistency and coherence across an organization's architecture.
- **Stakeholder Management:** Engaging stakeholders effectively is crucial for architectural success. The TOGAF Standard provides methods for identifying, analyzing, and managing stakeholders.
- **Architecture Patterns:** Architecture patterns provide standardized solutions to common architectural problems.
- **Gap Analysis:** Gap analysis is a technique used to identify differences between the Baseline and Target Architectures.
- **Interoperability Requirements:** Describes a technique for determining interoperability requirements.
- **Business Transformation Readiness Assessment:** A technique for identifying business transformation issues.
- **Risk Management:** Risk Management ensures that potential issues are identified and mitigated before impacting the architecture.
- **Architecture Alternatives and Trade-Offs:** Identify alternative Target Architectures and perform trade-offs between the alternatives.

The eight techniques mentioned above are detailed in the following sections.

14.2. Principles

In the TOGAF Standard, principles play a crucial role in guiding decision-making and ensuring alignment between business objectives and strategies. Principles are often established within different domains and at different levels of an organization. Although the use of principles is not really a technique (rather a necessity), the TOGAF Standard classifies their creation and development as such.

The TOGAF Standard distinguishes between *Enterprise Principles* and *Architecture Principles*, each serving different purposes. These are the two key domains that inform the development and utilization of architecture:

- **Enterprise Principles:** High-level, general guidelines that shape the overall strategy, culture, and decision-making framework of an organization. These principles apply broadly across the entire organization, influencing business operations, IT strategy, and governance. They typically address aspects such as business agility, innovation, customer focus, security, compliance, and sustainability.
- **Architecture Principles:** More specific guidelines that direct the design and development of Enterprise Architecture. They apply to architecture development efforts, ensuring consistency, scalability, and alignment with business goals. These principles provide direction for technology choices, system integration, and IT governance.

The key differences between Enterprise Principles and Architecture Principles are outlined in Table 14-1.

Table 14-1. *Differences between Enterprise and Architecture Principles*

Aspect	Enterprise Principles	Architecture Principles
Scope	Organization-wide, applies to business and IT	Focused on Enterprise Architecture
Purpose	Guides overall decision-making and strategy	Provides a framework for architecture design and execution
Application	Broad, influences policies and culture	More technical, used in architecture development
Examples	Agility, customer focus, security	Reusability, scalability, standardization

14.2.1 The Added Value of Principles

Principles, in general, provide clarity, consistency, and direction in decision-making. Their value lies in ensuring that business and IT strategies support the same objectives. In other words, *principles guide strategic alignment*. They also allow for standardizing approaches across different teams, reducing conflicts and inefficiencies. Principles enable better decision-making by providing a structured framework to evaluate options and trade-offs. They help establish clear policies and accountability in architecture development. Last but not least, principles ensure a shared understanding of organization and architecture goals and objectives.

Within an Enterprise Architecture approach, principles serve as a foundation for designing flexible and scalable architectures that align with the aforementioned business goals and objectives. They ensure compliance with regulations and industry standards and are able to

reduce complexity and duplication by enforcing standardization and interoperability.

14.2.2 The Structure of a Principle

By defining principles clearly and applying them consistently, organizations can create a well-structured, effective Enterprise Architecture that supports both short-term and long-term goals.

Each principle in the TOGAF Standard is typically structured as shown in Table 14-2.

Table 14-2. *Structure of a principle*

Structure Element	Description
Name	A short, clear title.
Statement	A concise declaration of the principle's intent.
Rationale	The justification for the principle, explaining why it is important.
Implications	The practical impact of the principle, including potential trade-offs and required actions.

There are some essential elements that a principle must incorporate. Table 14-2 lists four structural elements that every principle must contain:

Name: The key is to choose a short title that leaves nothing to the imagination. The best result is achieved by using the core essence of the principle. To ensure the applicability of the principle, it is also wise to avoid using specific product names. Also, the use of words that can be understood in more than one way is out of the question.

Statement: The structure element statement should clearly and concisely convey the basic rule without ambiguity. The intent of the principle is often used for this purpose.

Rationale: With the rationale, the added business value of adhering to the principle is explained. Often, consistency with other principles is also mentioned. If a particular principle takes precedence over another principle, that is a reason to describe it in the rationale. This structure element describes the justification for the principle. It explains why adherence to the principle is important.

Implications: This structure element identifies the requirements for applying the principle. This is usually done by expressing the requirements in terms of resources, time, money, or specific activities required.
The impact of a principle after it has been adopted should be clearly articulated, including potential trade-offs and required actions. In effect, the implications answer the question, "How does this affect the business?"

In addition to using the structural elements mentioned above (name, statement, rationale, and implications), it is important to ensure that principles meet a number of quality requirements. Principles are used in decision-making, and for that reason alone it is wise to make them as clear as possible. The following five criteria distinguish a good set of principles [20]:

Understandable: The underlying tenets can be quickly grasped and understood by individuals throughout the organization. The intent of the principle is clear and unambiguous, minimizing violations, whether intentional or unintentional.

Robust: Principles should enable high-quality decisions about architectures and plans and create enforceable policies and standards. Each principle should be sufficiently definitive and precise to support consistent decision-making in complex, potentially controversial situations.

Complete: Every potentially important principle for managing information and technology for the organization is defined – the principles cover every conceivable situation.

Consistent: Strict adherence to one principle may require a loose interpretation of another principle. The set of principles must be expressed in a way that allows for a balance of interpretations. Principles should

not be so contradictory that adherence to one principle would violate the spirit of another. Each word in a statement of principles should be carefully chosen to allow for consistent yet flexible interpretation.

Stable: Principles should be enduring, yet able to accommodate change. An amendment process should be established to add, delete, or modify principles after their initial ratification.

Enterprise and Architecture Principles are stored in the Governance Repository section of the Architecture Repository, as shown in Figure 14-1.

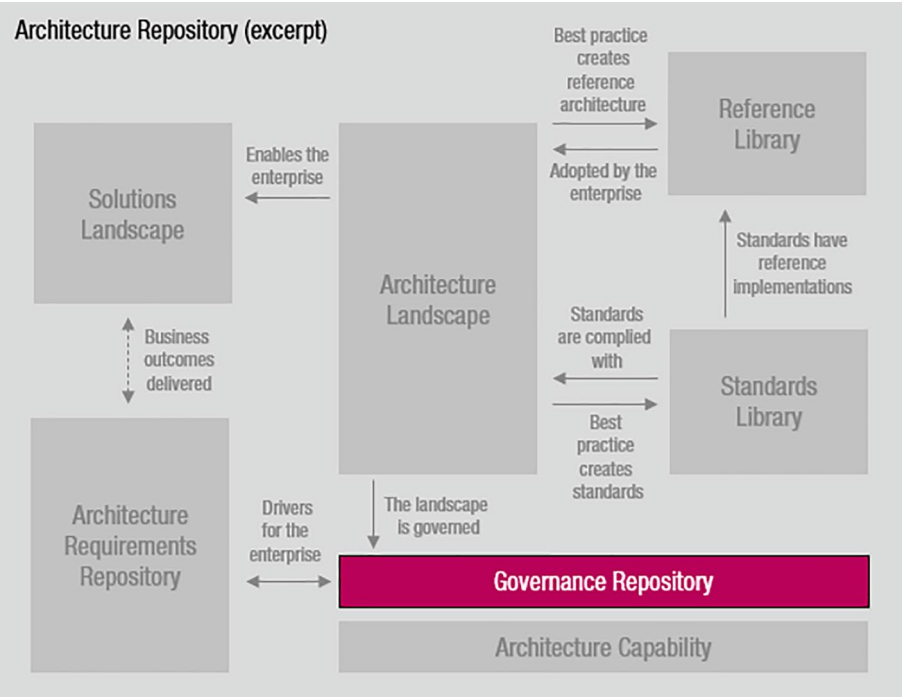


Figure 14-1. *Excerpt of the Architecture Repository, showing Enterprise and Architecture Principles stored in the Governance Repository*

The Series Guide "Using the TOGAF Standard in the Digital Enterprise" uses the term Enterprise Architecture Principles [23].

With this, the TOGAF Standard seems to introduce a third variant of the entity principle.

In addition to Enterprise and Architecture Principles – which are very different in form and application – the term *Enterprise Architecture Principles* actually refers to Architecture Principles.

This inconsistency in the Series Guide seems to inadvertently merge Enterprise Principles and Architecture Principles when they should be distinct.

Enterprise and Architecture Principles are different and should not be confused. They should certainly not be considered one and the same. However, it is common for an organization to simply talk about *principles*. Nine times out of ten, these principles (which sometimes resemble requirements) come from IT. Most other departments in an organization are unaware of the existence of principles or do not apply them.

14.2.3 Enterprise Principles

Enterprise Principles are high-level, general guidelines that shape the overall strategy, culture, and decision-making framework of an organization. This type of principles ensures that decisions in an organization are made in harmony. They are an essential part of architecture governance.

Given the primarily strategic nature of Enterprise Principles, they would ideally be drafted by the board of directors or another senior management body within the organization. However, the reality is that they are initially created by the Enterprise Architect and later approved

by the board of directors (or the aforementioned senior management). The buy-in of all stakeholders involved in the creation of the Enterprise Principles is necessary to provide the required support.

Enterprise Principles provide a basis for decision-making throughout an enterprise and inform how the organization sets about fulfilling its mission [20].

Because Enterprise Principles (see Table 14-3) are described at a high level, it is quite common for several of these types of principles to exist in an organization. For example, it is not inconceivable for different departments to have their own set of principles. Of course, it is important that all Enterprise Principles are aligned to avoid overlap or contradiction between them.

Table 14-3. *Example of an Enterprise Principle*

Name	Statement	Rationale	Implications
Comply with laws and regulations.	Information management processes comply with all relevant laws, policies, and regulations.	To comply with laws and regulations, relevant laws and regulations are considered in the development, procurement, and design of systems, products, and services.	• The organization must be mindful to comply with laws, regulations, and external policies regarding the collection, retention, and management of data. • Efficiency, need, and common sense are not the only drivers. Changes in the law and changes in regulations may drive changes in our processes or applications.

The Principles Catalog artifact shown in Table 12-5 in Chapter 12, Section 12.3.2, lists additional Enterprise Principles.

Rooted in business fundamentals, Enterprise Principles play a pivotal role in shaping the foundation for effective architecture governance. After thoroughly understanding the organizational context, it's crucial to define a tailored set of Enterprise Principles that aligns with the organization's unique needs. This ensures a strong, adaptable framework that supports both governance and strategy.

14.2.4 Architecture Principles

Architecture Principles are intended to provide guidance for the architecture process. They also provide direction for the development, maintenance, and use of the Enterprise Architecture. These principles are drawn up by the Enterprise Architect in collaboration with key stakeholders. They are formally approved by the Architecture Board.

Architecture Principles are a set of principles that relate to architecture work [20].

What is often seen in organizations is the existence of one or more sets of principles. These sets then often form a hierarchy. For example, it is quite common to have multiple Architecture Principles for each segment of the architecture.

Figure 14-2 shows how these *segment principles* (derived from segments such as business, data, application, or technology domains) are collectively informed by Enterprise Principles. Architecture Principles build on what has been established at the high level. In this way, the Enterprise Principles define the boundaries for the Architecture Principles.

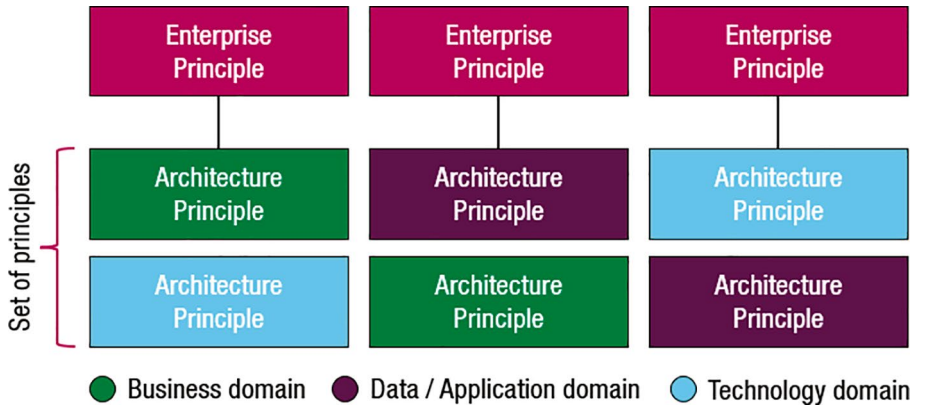


Figure 14-2. *Sets of Architecture Principles that refer to specific segments*

Table 14-4. *Example of an Architecture Principle*

Name	Statement	Rationale	Implications
Common vocabulary and definitions	Data is defined consistently throughout the organization, and the definitions are understandable and available to its users.	The data that will be used in the development of applications must have a common definition.	• The organization must establish the initial common vocabulary for the business; the definitions will be used uniformly throughout the organization. • Whenever a new definition is required, the effort will be coordinated and reconciled with the corporate glossary of descriptions [20].

Architecture Principles (see Table 14-4 for an example) are fundamental guidelines that dictate the use and deployment of resources and assets throughout an organization. They represent a consensus among various organizational elements and serve as a foundation for future decision-making. Each Architecture Principle should be explicitly linked to the organization's business goals and key architectural drivers.

14.3. Stakeholder Management

Being able to communicate properly with stakeholders – the key players within the organization – is essential. To identify these key players in an organization, *Stakeholder Management* can be applied. Stakeholder Management is the process of identifying and understanding the needs, interests, and expectations of an organization's various stakeholders and planning and implementing actions to address those interests and improve cooperation and communication with them [20].

The TOGAF Standard provides a structured approach to Stakeholder Management, ensuring that architectural projects align with organizational objectives and gain the necessary support from all involved parties.

The stakeholder concept is fundamental to representing an individual, group, or organization that has an interest in the outcome of an architecture effort. Stakeholders are critical to the success of any Enterprise Architecture initiative because they provide input, have concerns, and influence decision-making throughout the architecture development process. The stakeholder concept is used to *identify* and *analyze* their *concerns* and *interests*. Understanding stakeholder needs and expectations helps define architecture requirements and ensures that the architecture effectively addresses business goals.

Stakeholders are actively involved throughout the architecture development process. Enterprise Architects work with stakeholders to

gather information, obtain feedback, and validate architectural decisions. Effective stakeholder engagement results in a shared understanding of business goals and ensures buy-in for the proposed architecture.

The stakeholder concept facilitates effective communication with various groups within the organization. Architects use the concept to document stakeholder relationships, communication channels, and key roles within the architecture effort. It helps manage expectations and ensure that stakeholders are kept informed of progress and results.

Stakeholders also play a critical role in assessing the impact of architecture decisions on various business areas. By understanding their perspectives (also called views; see Chapter 7, Section 7.6), architects can identify potential risks and opportunities associated with architectural changes and align the architecture with the organization's strategy. The primary goal of Enterprise Architecture is to align all the architecture work originating from the architecture domains and segments with an organization's goals. Understanding stakeholder perspectives helps architects prioritize initiatives and allocate resources to projects that deliver the most value to the organization and its stakeholders.

By interacting with stakeholders, architects can elicit and document their needs, which become the basis for developing architectural models and solutions. Stakeholders are an important source of requirements for the architecture. The stakeholder concept helps architects make informed decisions. By considering the interests and concerns of different stakeholders, architects can balance conflicting requirements and ensure that architectural decisions are well accepted and aligned with business priorities.

The stakeholder concept is closely related to governance in Enterprise Architecture. Understanding who the stakeholders are and what their interests are helps architects establish governance mechanisms to address their concerns, maintain transparency, and ensure accountability.

In the context of the TOGAF Standard, a stakeholder is any individual, group, or organization that can affect, be affected by, or perceive

themselves to be affected by an Enterprise Architecture initiative. Effective Stakeholder Management involves identifying these parties, understanding their concerns and interests, and engaging with them appropriately to ensure the success of the architectural endeavor.

Early identification of influential or key stakeholders allows their input to shape the architecture, ensuring support and enhancing the quality of the models produced.

Gaining support from powerful stakeholders can help secure necessary resources, increasing the likelihood of project success. Knowing who the stakeholders are ensures that the architecture and related work can be communicated and aligned. It is therefore essential to identify them as early as possible and to understand their needs and expectations. Analysis of the identified stakeholders will help to paint this picture.

Engaging stakeholders early and often ensures that they understand the architecture process and its benefits and can more actively support it when needed. This can be achieved through effective communication. A *Communications Plan* will be of great help in achieving this. Such a plan is often used to ensure effective communication of the necessary information to the relevant stakeholders at the right time.

In order for an architecture team to anticipate reactions from stakeholders, it is important to understand their perspectives. This will help in planning actions to leverage positive responses while addressing negative ones. Identifying conflicting objectives among stakeholders early enables the development of strategies to resolve arising issues.

The TOGAF Standard recommends incorporating stakeholder analysis during Phase A (Architecture Vision) of the Architecture Development Method and updating it throughout each following phase. This form of continuous engagement ensures that new stakeholders are identified and their concerns are addressed as the project progresses through phases such as Phase E (Opportunities and Solutions), Phase F (Migration Planning), and Phase H (Architecture Change Management).

A practical approach to Stakeholder Management is described in the step-by-step guide outlined below.



Identify Stakeholders: Knowing who the stakeholders are ensures that the architecture and related work can be communicated and aligned.

Analyze and Classify Stakeholder Positions: Stakeholders need to be analyzed to understand their needs and expectations.

Plan Communications: Creating a Communications Plan ensures effective communication of the necessary information to the relevant stakeholders at the right time.

Engage Stakeholders: Explore options for additional, more effective ways and means of communication.

Monitor and Review: Evaluate the effectiveness of stakeholder interactions on a regular basis. If *necessary*, adjust the stakeholder engagement and communication strategies.

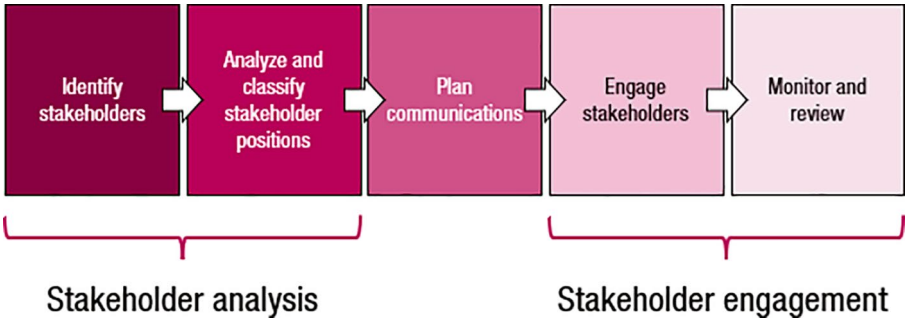


Figure 14-3. *Steps to stakeholder analysis and stakeholder engagement*

The first two steps, *identify stakeholders* and *analyze and classify stakeholder positions*, are part of the overall stakeholder analysis process, as shown in Figure 14-3. The step called *plan communications* is all about creating a Communications Plan. Finally, the last two steps, *engage stakeholders* and *monitor and review*, are part of stakeholder engagement.

14.3.1 Stakeholder Analysis

The Architecture Vision Phase details the process of identifying the stakeholders and determining their requirements. Knowing who the stakeholders are (this includes both organizations and individuals within those organizations) ensures that the architecture and related work can be communicated and aligned. A critical part of communicating with stakeholders is creating an effective Communications Plan.

Having a clear view of the stakeholders and their requirements and concerns is an important prerequisite for creating the Solution Concept Diagram. This diagram is based on the organization's goals and objectives and stakeholder requirements and constraints.

After identifying stakeholders, it is imperative to *analyze* these groups to understand their needs and expectations. This helps to prioritize their requirements and determine the best way to design the architecture to meet their needs. In order to achieve this, a Stakeholder Analysis Model (see Table 14-5) is created.

Table 14-5. *Stakeholder Analysis Model*

Stakeholder Group	Stakeholder	Ability to Disrupt Change	Current Understanding	Required Understanding	Current Commitment	Required Commitment	Required Support
CxO	Maria Summers (CEO)	H	H	H	M	H	H
	Chris West (CIO)	H	M	H	L	M	H
	James Noble (CFO)	M	M	M	L	M	M
HR Management	Anna Carter (Manager HR)	L	L	M	L	M	M

The Stakeholder Analysis Model includes the stakeholder group, the specific stakeholder, and the stakeholder's ability to disrupt change. The latter is a reference to the level of power a particular stakeholder has within the organization. It illustrates the likelihood that a particular stakeholder can disrupt or actively promote the architecture work.

A stakeholder's current understanding of and commitment to the architecture work are also captured in the model. The level of commitment can be compared to the level of interest mentioned in the Stakeholder Power Grid (see Figure 14-4). Finally, the required levels of understanding, commitment, and support are identified.

A Stakeholder Power Grid is used to determine the power and interest levels of the stakeholders captured in the Stakeholder Analysis Model. The Stakeholder Power Grid allows you to assign a classification to each stakeholder. The grid consists of four quadrants, each representing one of the following four types of classifications:

Key Players: This classification indicates that these are stakeholders with a high level of interest and power within the organization (*high power/high interest*). Therefore, the frequency of communication to this group of stakeholders will be the highest. An example of this stakeholder classification is data and process owners stakeholder group in Table 14-8.

Keep Satisfied: Stakeholders classified as keep satisfied have the same level of power in the organization as the key players, but their level of interest is lower (*high power/low interest*). Therefore, this group can be informed with a slightly lower frequency. Most C-level positions in an organization fall into this classification.

Keep Informed: The level of interest among this group of stakeholders is high, but the power they can exercise within the organization is lower (*low power/high interest*). Nevertheless, it is wise to keep this group informed on a regular basis. For this group, the choice is between informing them by email and in the form of a meeting (physical or digital). An example of a stakeholder in this classification is the program manager.

Minimal Effort: Although this term has a negative connotation, this is far from the intention. It refers to the fact that stakeholders in this group have low levels of both interest and power. However, this group can still contain stakeholders that are very important to the architecture work. The employees group mentioned in Table 14-8 falls into this classification.

Using a Stakeholder Power Grid (see Figure 14-4), the architect can get a good understanding of each stakeholder's role, their level of influence, and their interest in the project. These levels are visualized in Figure 14-4 as *level of power* and *level of interest,* respectively.

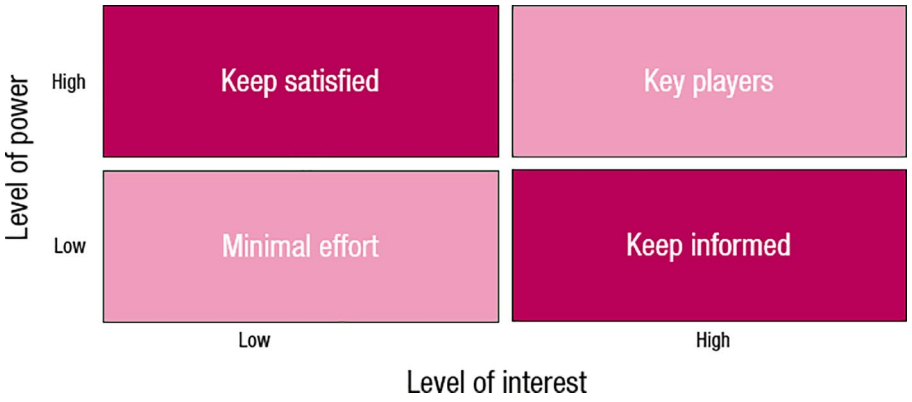


Figure 14-4. *Stakeholder Power Grid*

The stakeholders that were analyzed at the beginning of step 2 (see Figure 14-3) are classified using the Stakeholder Power Grid. Their power and interest levels, as well as their classification, are recorded in the Stakeholder Analysis Model (see Table 14-6).

Table 14-6. *Stakeholder Analysis Model with Stakeholder Classification*

Stakeholder Group	Stakeholder	…	Power	Interest	Classification
CxO	Maria Summers (CEO)	…	H	L	Keep satisfied
	Chris West (CIO)	…	H	H	Key player
	James Noble (CFO)	…	L	H	Keep informed
HR Management	Anna Carter (Manager HR)	…	L	L	Minimal effort

Each classification comes with a set of requirements regarding the stakeholder's communication needs. For example, it contains information about the appropriate frequency of communication and the most appropriate method of communication. This information forms the basis for the Communications Plan.

14.3.2 Stakeholder Mapping

Another essential Stakeholder Management tool is the creation of a Stakeholder Catalog. This map contains not only the stakeholders and their classification, but also the *key concerns* of the stakeholders. These detail their primary areas of focus. The Stakeholder Catalog (see Table 14-7) also contains references to the various catalogs, matrices, diagrams, and maps that are of interest to a particular (group of) stakeholder(s).

Table 14-7. *Stakeholder Catalog*

Stakeholder	Key Concerns	Classification	Catalogs, Matrices, Diagrams, and Maps
Maria Summers (CEO)	The high-level drivers, goals, and objectives of the organization and how these are translated into an effective process to advance the business	Keep satisfied	Strategy/Goal Matrix Goal/Objective Matrix
Chris West (CIO)	Alignment of IT strategy with business goals, optimizing IT costs, and supporting business agility through technology	Key player	Goal/Objective Matrix Objective/Initiative Matrix
James Noble (CFO)	Budget efficiency and cost control in organizational projects, long-term financial sustainability of architecture decisions	Keep informed	Application Portfolio Catalog Technology Portfolio Catalog
Anna Carter (Manager HR)	Supporting workforce needs in alignment with organizational transformation, ensuring employee adoption of new processes and technologies	Minimal effort	Business Process Map Organization/Business Process Matrix

A completed Stakeholder Catalog is an important resource that can be used frequently. The stakeholders identified in the Stakeholder Catalog need to be approached and informed on a regular basis. A Communications Plan is used for this purpose.

14.3.3 Communications Plan

Effective stakeholder communication is critical to keeping all stakeholders sufficiently invested in the architecture work. One way to achieve this is to create a Communications Plan. Such a plan ensures effective communication of the necessary information to the relevant stakeholders at the right time. It is therefore one of the critical success factors (CSFs) for a sound implementation of an Enterprise Architecture [11].

The Communications Plan should describe the communication needs of each stakeholder group, such as the information they need, the frequency of communication, and the methods for sharing that information. It should also state the purpose of the communication to the stakeholders. Of course, this will not be the same for each stakeholder group, and this distinction needs to be well-defined.

The Stakeholder Power Grid shown in Figure 14-4 illustrates how stakeholders can be classified. With each classification comes a set of requirements regarding the communication needs of the stakeholders. It is important to include the moments – especially the frequency of these moments – at which stakeholders are informed or updated.

The TOGAF Standard indicates that the scope of a Communications Plan is limited to key stakeholders. In my view, communication does not stop there.

It is precisely by communicating to the entire organization what the expected impact of implementing the Enterprise Architecture will be that everyone becomes involved. It becomes more of everyone, rather

than just those directly involved. This strengthens the acceptance of the Enterprise Architecture as well as working with architecture.

There are two components that are essential to include in a Communications Plan.

The first is to include and interpret what everyone's role will be in implementing the Enterprise Architecture. This does not need to be detailed; an overall articulation in terms of roles to be fulfilled is sufficient. If people can see themselves in the roles outlined, with the associated roles and responsibilities, it will be easier to buy into the implementation. Everyone likes to make a meaningful contribution. What is sometimes missing in practice is the interpretation of that meaning. Once the meaning is unclear, people's enthusiasm for making the desired contribution diminishes. So be clear and precise in your descriptions.

The second essential part of the Communications Plan is to schedule consultation moments when the organization as a whole is updated or informed about the progress of the implementation and the method(s) to be used. Transparency in the approach is very important.

Of course, this does not always mean that all employees are informed at the same time; it is certainly possible to create a division into small(er) groups.

Once the Communications Plan has been created, the architect must implement it to foster effective communication and collaboration. From then on, stakeholders are regularly updated on the progress of the implementation, and feedback sessions are offered as opportunities for stakeholders to provide input.

Table 14-8. *Communications Plan*

Stakeholder Group	Classification	Communication Goals	Communication Method	Communication Frequency
CxO	Keep satisfied	Provide insight into how the high-level drivers, goals, and objectives are translated into architecture work.	Meeting	Every fortnight
Data and process owners	Key players	Ensure the consistent use and governance of the organization's business, data, application, and technology assets.	Meeting	Monthly
Program manager	Keep informed	On-time, on-budget delivery of change initiatives that will lead to the implementation of the architecture.	Meeting	Monthly

(*continued*)

Table 14-8. (*continued*)

Stakeholder Group	Classification	Communication Goals	Communication Method	Communication Frequency
Employees	Minimal effort	Assess the impact of the implementation of the Enterprise Architecture with regard to the day-to-day duties and tasks.	Email message, interspersed with meeting	Monthly
Human resources	Minimal effort	Ensure that the roles and actors who are required to support the implementation of the architecture are available.	Email message	Quarterly

A standard Communications Plan (Table 14-8) often looks static and business-like and does not usually provide the most effective means of communication in practice. It is therefore advisable to explore options for additional means of communication in order to enhance stakeholder engagement.

14.3.4 Stakeholder Engagement

A more effective way of communicating might be to use appealing and sophisticated visuals and designs that can be used to capture and hold the attention of stakeholders. Common forms of this are presentations and infographics, but posters or specially decorated rooms where the information is highlighted can also help. Encourage stakeholders to provide input and express concerns to foster a collaborative environment.

Assess stakeholder engagement by regularly evaluating the effectiveness of stakeholder interactions. If necessary, adjust the stakeholder engagement and communication strategies. Also update the Stakeholder Analysis Model on a regular basis. When new stakeholders emerge (e.g., new employees are joining the organization), it is advisable to revise stakeholder information and the accompanying engagement plans to reflect any changes in stakeholder positions or project dynamics.

Effective Stakeholder Management, as outlined in the TOGAF Standard, is essential for the success of Enterprise Architecture initiatives. By systematically identifying, analyzing, and engaging stakeholders, organizations can ensure that architectural projects are aligned with business objectives, gain necessary support, and address concerns proactively. Implementing the step-by-step plan detailed above will facilitate structured Stakeholder Management, contributing to the overall success of the architecture development process.



Further Reading

For more information on Stakeholder Management, refer to *The TOGAF Standard, 10th Edition*, document number C220.

14.4. Architecture Patterns

Chapter 7, Section 7.5.2, discussed building blocks. They are considered the backbone of the architecture and serve as essential, reusable components in architectural solutions. Building blocks form a modular foundation that is designed to be generic and adaptable, ensuring that they can be seamlessly integrated with other building blocks. Their inherent reusability allows architects to combine them in different configurations, promoting efficiency and scalability in the development of architectural solutions.

Architecture patterns, on the other hand, put building blocks into context. They use building blocks to create a reusable solution to a problem.

Building blocks are what you use; patterns can tell you how you use them, when, why, and what trade-offs you have to make in doing so [20].

Architecture patterns offer architects a way to use combinations of Architecture and Solution Building Blocks that have proven to deliver an effective solution in the past.

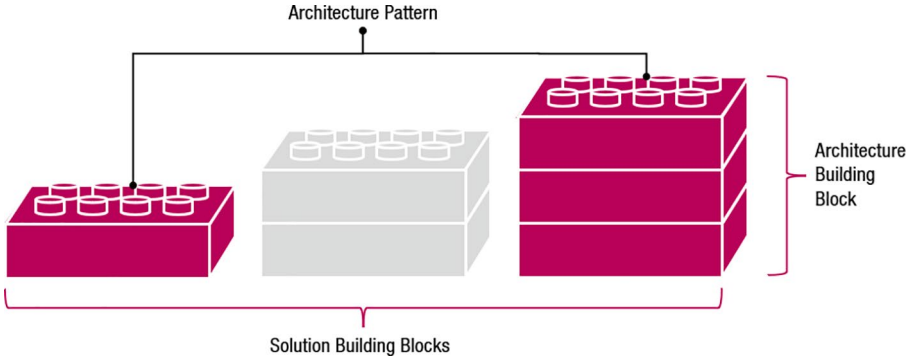


Figure 14-5. *Architecture patterns and building blocks*

As shown in Figure 14-5, architecture patterns are composed of Architecture and Solution Building Blocks. The figure illustrates a set of *Solution Building Blocks* (horizontal) and *Architecture Building Blocks* (vertical). An architecture pattern can be a *combination* of one or more Solution Building Blocks and one or more Architecture Building Blocks.

In everyday practice, when the word *pattern* is used, it is still common to think of software design patterns. This is because many people in the software field use the term architecture to refer to software. For this reason, an architecture pattern is often confused with a design pattern.

The TOGAF Standard has not yet formally incorporated the use of architecture patterns into the Architecture Development Method.

Architecture patterns have not (as yet) been integrated into the TOGAF Standard [20].

However, it can be stated that during the execution of phases B, C, and D – the moment when reusable building blocks are identified – architecture patterns can also be defined.

14.5. Gap Analysis

A gap analysis is a process that identifies the difference between an organization's Baseline and Target Architectures in terms of its capabilities, processes, applications, and technology. The purpose of a gap analysis is to identify areas where improvements are needed to achieve the desired future state.

The basic premise is to highlight a shortfall between the Baseline Architecture and the Target Architecture, that is, items that have been deliberately omitted, accidentally left out, or not yet defined [20].

There are a few basic steps to perform a gap analysis:

- Identify Baseline and Target Architectures.
- Compare the two states to highlight gaps.
- Categorize gaps as missing functionalities, technology gaps, or process inefficiencies.
- Create work packages of grouped activities to fill the gaps.
- Prioritize work packages based on business impact and feasibility.

The results of the gap analysis are used to inform subsequent phases of the Architecture Development Method, such as phases B, C, and D, where the detailed architecture is developed based on the architectural vision.

Remember the fictional hospital described in Chapter 5 (Section 5.1.1)?

This particular hospital concluded earlier on that it lacked a required capability, Inflow Support, based on the Business Capability Heat Map in Figure 12-9 (Chapter 12, Section 12.5.2).

To automate the missing capability later, a new application (HR Recruitment Application B) would have to be implemented. Table 12-57, the Consolidated Gaps, Solutions, and Dependencies Matrix, lists the implementation of the new application as a possible solution for automating the capability.

Figure 14-6 shows one way to identify and compare the Baseline and Target Architectures by highlighting the differences between them. The figure shows the identified capabilities and applications that exist either in both the Baseline and the Target Architectures, only in the Baseline Architecture, or only in the Target Architecture.

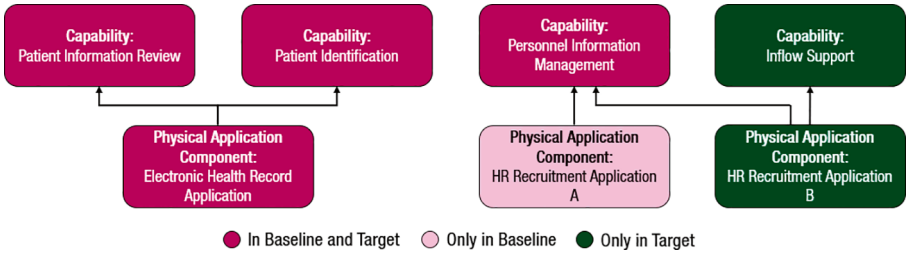


Figure 14-6. *Identification of the Baseline and Target Architectures*

The figure illustrates that HR Recruitment Application A exists in the Baseline Architecture, but is replaced by HR Recruitment Application B in the Target Architecture. After the new application is implemented, it enables the automation of the Inflow Support capability, which does not exist in the Baseline Architecture. Due to the implementation of the new application, the Personnel Information Management capability is also automated by HR Recruitment Application B.

Figure 14-7 shows the phases of the Architecture Development Method in which the gap analysis steps take place. Most of the work is done in phases B, C, and D. This is where the gaps between the Baseline and Target Architectures are identified. Then, in Phase E (Opportunities and Solutions), the real analysis takes place. The necessary changes to the architecture are identified, and the activities required to close the gaps are grouped into work packages.

The figure shows three architecture states at the top row: Baseline, Transition, and Target. Right below that are two gaps visible. Gap A illustrates the gap between the Baseline and Transition Architectures, and Gap B illustrates the gap between the Transition and Target Architectures. The bottom row shows two work packages. The one on the left contains the activities necessary to close Gap A, and the other one contains activities to close Gap B.

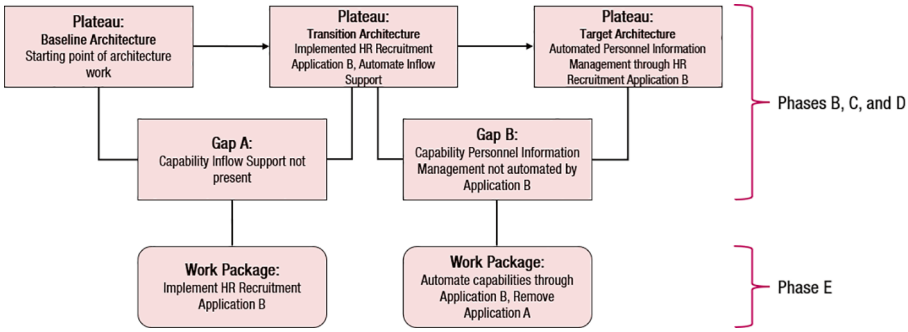


Figure 14-7. *Gap analysis during phases B–E of the ADM*

As shown in Figure 14-7, the gaps between the Baseline and Transition Architectures (Gap A) on the one hand and the Transition and Target Architectures (Gap B) on the other are identified. Gap A shows that the capability Inflow Support is not yet present. A work package with activities is created to close the gap. In this case, the work package contains the activity to implement HR Recruitment Application B, which will eventually automate the capability Inflow Support.

Gap B, which details the gap between the Transition and Target Architectures, shows that the Personnel Information Management capability is not yet automated by HR Recruitment Application B. Again, a work package is created containing activities to automate the Personnel Information Management and Inflow Support capabilities through HR Recruitment Application B and to remove HR Recruitment Application A from the architecture.

It is recommended that the actions listed above be described in a Gap Analysis Matrix (see Table 14-9).



Set Up the Matrix: Begin by drawing a grid. Label the rows with the Architecture Building Blocks from the Baseline Architecture. Label the columns with the Architecture Building Blocks from the Target Architecture.

Add New and Eliminated Sections: Extend the matrix by adding an extra row at the bottom labeled *New* and an extra column on the right labeled *Eliminated*. This will help track additions and removals.

Identify Common Building Blocks: For each Architecture Building Block that exists in both the Baseline and Target Architectures, mark the intersecting cell in the matrix with *Included*. This shows which components remain unchanged.

Review Missing Building Blocks: Look for Architecture Building Blocks that are present in the Baseline Architecture but absent in the Target Architecture. These will align with the *Eliminated* column. Assess whether removing these components is appropriate and document the reasoning.

Document New Building Blocks: Identify Architecture Building Blocks that appear in the Target Architecture but not in the Baseline. Place these in the *New* row of the matrix. Consider how these additions align with the architectural goals and document any necessary implementation steps.

If an Architecture Building Block was correctly eliminated, mark it as such in the appropriate *Eliminated* cell. If it was not, this means that an accidental omission in the Target Architecture has been uncovered that must be addressed by reinstating the Architecture Building Block in the next iteration of the architecture design. Mark it as such in the appropriate *Eliminated* cell.

Table 14-9. *Example Gap Analysis Matrix*

	Target Architecture				
Baseline Architecture ▼	Patient Information Review	Patient Identification	Personnel Information Management	Inflow Support	Eliminated
Patient Information Review	Included				
Patient Identification		Included			
Personnel Information Management			Included		HR Recruitment Application A
New			Gap: HR Recruitment Application B	Gap: HR Recruitment Application B	

When the exercise is completed, anything under *Eliminated* or *New* is a gap that should either be explained as correctly eliminated or marked as to be addressed by performing activities to reinstate or develop/procure the building block. Work packages are used to group these activities. The work packages find their way to the Architecture Roadmap, which is described in Chapter 12, Section 12.8.3.

14.6. Interoperability Requirements

The idea behind the interoperability technique is to compare a set of entities and determine what form and degree of interoperability is required between them. This involves determining the degree of interoperability needed to share information and services.

For example, interoperability may be required between two stakeholders or between a stakeholder and an application. Cross-mapping between the two entities provides a picture of what can exchange information with whom and in what manner. The cross-mapping technique is also used to determine the requirements for the desired interoperability.

A definition of interoperability is "the ability to share information and services". Defining the degree to which the information and services are to be shared is a very useful architectural requirement, especially in a complex organization and/or extended enterprise [20].

The TOGAF Standard recommends using the cross-mappings just described to determine the *interoperability requirements*. An example is shown in Table 14-10.

Table 14-10. *Interoperability Requirements between stakeholders*

Stakeholders ▶ / ▼	A	B	C	D	E
A		2	3	2	3
B	2		3	2	3
C	3	3		2	2

Table 14-10 shows a cross-mapping between stakeholders (A–E) and lists the associated degree of interoperability. Since this particular table contains the entity *stakeholder*, it is created from a Business Architecture perspective. Therefore, it is developed during the corresponding phase (Phase B) of the Architecture Development Method. The content of a cross-mapping is related to the phase of the Architecture Development Method in which it is created.

Table 14-10 hints at the fact that the TOGAF Standard distinguishes four degrees (levels) of interoperability. These levels are indicated by numbers:

- **Degree 1 (Unstructured Data Exchange):** Involves the exchange of human-interpretable unstructured data, such as the free text found in operational estimates, analyses, and papers
- **Degree 2 (Structured Data Exchange):** Involves the exchange of human-interpretable structured data intended for manual and/or automated handling, but requires manual compilation, receipt, and/or message dispatch
- **Degree 3 (Seamless Sharing of Data):** Involves the automated exchange of data between systems based on a common exchange model
- **Degree 4 (Seamless Sharing of Information):** Is an extension of Degree 3 to the universal interpretation of information through data processing based on cooperating applications

Each of the aforementioned degrees can be further specified by assigning more specific descriptions to the degree. In this case, an alphabetic letter is added to the number of the degree. This is illustrated in Figure 14-8.

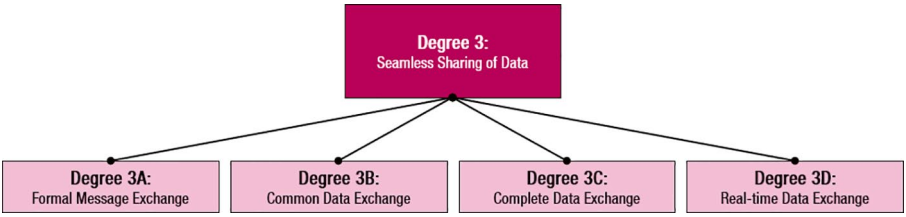


Figure 14-8. *Detailed degrees of interoperability*

Further detailing of the degrees can be used, for example, to express the level of standardization or compatibility of interoperability.

These degrees are very useful for specifying the way that information has to be exchanged between the various systems and provide critical direction to the projects implementing the systems.

In addition to cross-mapping between entities of the same type, it is also possible to compare different entities with each other. Table 14-11 shows an example of such a comparison between stakeholders on the one hand and applications on the other hand.

Table 14-11. *Cross-mapping of Interoperability Requirements between Stakeholder and Application Entities*

Applications Stakeholders ▼	CRM Application	Recruitment Application	Financial Application
Business Unit Managers	4D	3C	2A
HR Employees	3B	4D	–
Financial Controllers	3C	–	4D

The example in Table 14-11 shows that three applications were identified (CRM Application, HR Recruitment Application, Financial Application) as well as three stakeholder groups (Business Unit Managers, HR Employees, and Financial Controllers).

The cells in the table represent the degree of interoperability between each application and stakeholder. For example, the cell in *row 1*, *column 3,* of Table 14-11 represents the degree of interoperability between the *Financial Application* and *Business Unit Managers*. In this case, the degree of interoperability is *2A*, which represents *Structured Data Exchange* as the base degree. It also contains the letter B, indicating the detail level of *Common Data Exchange*. This means that the Financial Application can exchange data with Business Unit Managers in a structured format, but there may be some limitations or restrictions on the types of data that can be exchanged.

During the execution of the Architecture Development Method, the issue of interoperability arises several times. Figure 14-9 illustrates the integration of interoperability requirements into the Architecture Development Method. The Preliminary Phase has been removed for better readability.

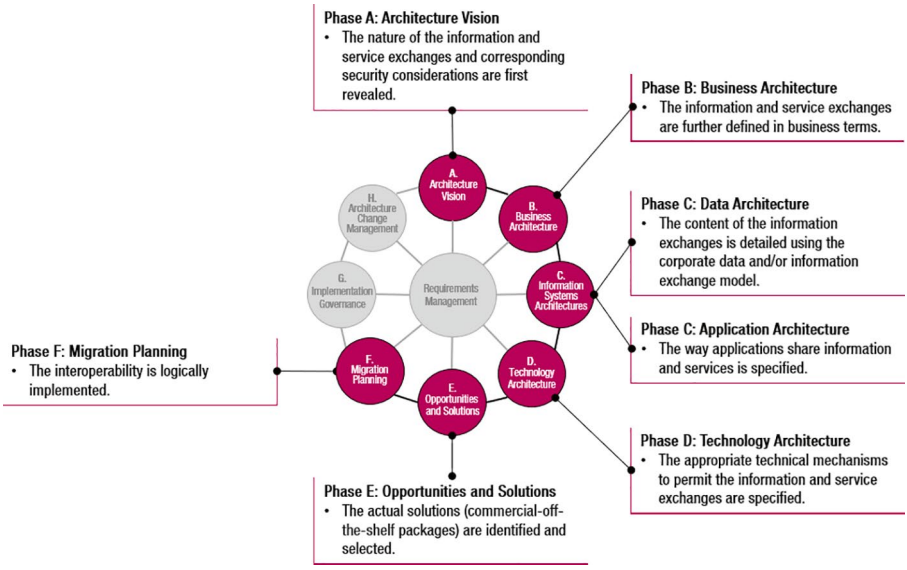


Figure 14-9. *Integration of interoperability requirements in the ADM*

One major advantage of applying the interoperability technique is *enhanced integration and communication between systems*. Interoperability ensures that different entities (such as people, systems, applications, and services) can effectively exchange and use information, reducing data silos and integration complexity.



Further Reading

For more information about interoperability requirements and how to apply them, refer to *The TOGAF Standard, 10th Edition*, document number C220.

The following website contains additional information about interoperability requirements and how to apply them:

https://guides.visual-paradigm.com/togaf-adm-top-10-techniques-part-6-interoperability-requirements/

14.7. Business Transformation Readiness Assessment

In the dynamic landscape of Enterprise Architecture, implementing an innovative architecture vision is not just a technical task. It is a profound organizational change that requires careful consideration of the human element. The TOGAF Standard recognizes the critical need to assess and address the organization's readiness for business transformation.

In the early phases of the Architecture Development Method, particularly Phase A (Architecture Vision) and the subsequent architecture definition phases (phases B–D), organizations conceptualize and design innovative solutions that often involve significant changes. These changes are not limited to technological shifts as they have a profound impact on the workforce, culture, and existing skills and competencies within the organization.

For example, the fact that a recruitment application will lead to a more effective way of managing candidate information and potentially reduce the number of staff required requires a keen understanding of the human resource implications. A change-averse culture and a workforce with narrow skill sets can potentially hinder the implementation of even the most well-designed architecture.

To ensure successful implementation of the architecture transformation in phases E and F, organizations must conduct a comprehensive readiness assessment. This is a collaborative effort between the organization's staff, particularly human resources, lines of business, and (IT) planners.

The steps in conducting a Business Transformation Readiness Assessment consist of identifying and defining the readiness factors that will significantly impact the organization during the transformation. These factors may include cultural readiness, workforce skills, leadership alignment, and change management capabilities.

Maturity models can be used to represent the readiness factors. They provide a structured framework for assessing the organization's Baseline and Target Architecture with respect to each readiness factor. The maturity model format described in Chapter 15 can be used as a template.

To assess the organization's readiness, a readiness factor rating must be determined. This assessment involves analyzing the current maturity level and identifying gaps that need to be addressed for a successful transformation.

Of course, the risks associated with each readiness factor also need to be identified and strategies developed to mitigate these risks. This includes understanding potential obstacles and challenges that could impede the transformation process. The Risk Management technique described in Section 14.8 can be used to accomplish this.

Finally, the results of the readiness assessment must be incorporated into the Implementation and Migration Plan created in Phase E and finalized in Phase F. This ensures that the organization's readiness for change is considered and addressed in the overall transformation strategy.

A Business Transformation Readiness Assessment table typically includes various factors that determine an organization's preparedness for a significant change. Table 14-12 shows an example of a Business Transformation Readiness Assessment.

Table 14-12. *Business Transformation Readiness Assessment*

Readiness Factor	Assessment Criteria	Score (1–5)
Leadership commitment	Level of commitment from top leadership	4
Employee engagement	Willingness and enthusiasm of employees for change	3
Organizational culture	Openness to new ideas and adaptability	5
Skills and competencies	Availability of required skills and competencies for the transformation	3
Risk Management	Identification and mitigation of potential risks	4
Stakeholder alignment	Alignment of stakeholders' interests and expectations	5

In the example presented in Table 14-12, each readiness factor is assessed on a scale of 1–5, with 5 being the highest readiness. The criteria can be customized based on the specific needs and nature of a business transformation.

Each readiness factor can be assigned a rating to score the factor based on its urgency, readiness status, and the degree of difficulty to remediate any issues (shown in Table 14-13). The urgency indicates how quickly action is needed, the readiness status provides an overall assessment of the current state (which refers to the score provided in Table 14-12), and the remediation difficulty represents the effort required to address any identified issues.

Table 14-13. *Business Transformation Readiness Assessment using a Readiness Factor Rating*

Readiness Factor	Urgency	Readiness Status	Remediation Difficulty
Leadership commitment	High	Good	Easy
Employee engagement	High	Acceptable	Moderate
Organizational culture	Moderate	Excellent	Moderate
Skills and competencies	Moderate	Good	Moderate
Risk Management	Moderate	Acceptable	Moderate
Stakeholder alignment	High	Excellent	Easy

The *readiness status* in Table 14-13 is based on the number that was assigned in the Business Transformation Readiness Assessment table (Table 14-12). The scores of 1–5 represent *poor*, *fair*, *acceptable*, good, and *excellent*, respectively.

Table 14-13 provides a quick overview for management, helping to understand the critical areas that require immediate attention and the overall readiness of the organization for the transformation initiative. The readiness factors, as part of an overall Implementation and Migration Plan, will have to be continuously monitored (Phase G) and rapid corrective actions taken through the Governance Framework to ensure that the defined architectures can be implemented.



Identify Key Readiness Factors: Begin by pinpointing the critical factors that influence the organization's ability to undergo transformation. These may include cultural readiness, workforce skills, leadership alignment, and change management capabilities. Understanding these elements is essential for a successful transformation journey.

Use Maturity Models for Assessment: Maturity models can be used to evaluate each readiness factor systematically. These models offer a structured framework to assess the organization's Baseline and Target Architecture concerning each factor.

Determine Readiness Factor Ratings: Analyze the current maturity levels of each readiness factor to identify gaps that need addressing. This evaluation helps in understanding the areas requiring improvement to facilitate a smooth transformation process.

Identify and Mitigate Associated Risks: Recognize potential risks linked to each readiness factor. Develop strategies to mitigate these risks by understanding possible roadblocks and challenges that could hinder the transformation. Proactive Risk Management ensures preparedness for unforeseen obstacles.

Integrate Findings Into Implementation and Migration Plan: Incorporate the insights gained from the readiness assessment into the Implementation and Migration Plan. This ensures that organizational preparedness is a core component of the overall transformation plan.

In the ever-evolving landscape of Enterprise Architecture, success hinges not only on the brilliance of solutions but also on the organization's ability to embrace and adapt to change. The TOGAF Standard's emphasis on performing a Business Transformation Readiness Assessment acknowledges the significance of the human element in the success of architectural endeavors.

By proactively identifying, evaluating, and mitigating the risks associated with organizational readiness, organizations can navigate the complexities of change with agility. A well-executed readiness assessment ensures that the human factors align harmoniously with the architectural vision, propelling the organization toward a future of innovation and resilience.



Further Reading

For more information about the Business Transformation Readiness Assessment, refer to *The TOGAF Standard, 10th Edition*, document number C220.

14.8. Risk Management

Risk Management is the second-to-last technique in the Architecture Development Method. Since it is addressed in several phases (Phase A and phases E–H), it can be seen as an integral part of architecture development. Applying Risk Management techniques ensures that risks are identified, assessed, and mitigated as part of the architecture development process.

According to the TOGAF Standard, Risk Management is specifically addressed in the following phases of the Architecture Development Method (see also Figure 14-10):

- **Phase A (Architecture Vision):** During this phase, an initial risk assessment is performed to identify high-level risks associated with the architecture work. This phase identifies significant risks that could affect the feasibility or success of the architecture initiative. The risks identified during Phase A are captured in the Statement of Architecture Work document.

- **Phase E (Opportunities and Solutions):** After the Baseline and Target Architectures have been created in phases B–D, the gaps discovered through the use of gap analysis allow for the discovery of risks related to the implementation of the Target Architecture. These implementation-related risks must be evaluated for potential architecture solutions. Phase E also evaluates risk mitigation strategies and ensures that risk considerations influence the selection of implementation approaches.
- **Phase F (Migration Planning):** The Consolidated Gaps, Solutions, and Dependencies Matrix (created in Phase E) serves as input for a detailed risk assessment related to migration and transition plans. Phase F uses Risk Management to identify risks standing in the way of a successful implementation and integration. It also ensures that mitigation plans and fallback strategies are in place.
- **Phase G (Implementation Governance):** The governance phase of the Architecture Development Method manages risks that arise during the implementation of the architecture. Managing risks from this phase ensures compliance with the architecture specification and governance standards. Phase G addresses emerging risks through governance oversight.
- **Phase H (Architecture Change Management):** Continuous risk monitoring is performed during Phase H with respect to changes to the architecture. The Architecture Change Management Phase identifies

risks associated with new requirements, technologies, or business changes. It implements risk response mechanisms as part of change management.

There are a few phases of the Architecture Development Method in which Risk Management is *not* explicitly addressed:

- **Preliminary Phase:** This phase focuses on architecture governance and Architecture Capability setup, but does not perform a risk assessment.
- **Phases B, C, and D:** Do not formally include Risk Management, although risks may be informally identified within the architectures. These risks, derived from the gap analysis performed during the development of the Baseline, Transition, and Target Architectures, are recorded in the Consolidated Gaps, Solutions, and Dependencies Matrix during Phase E.

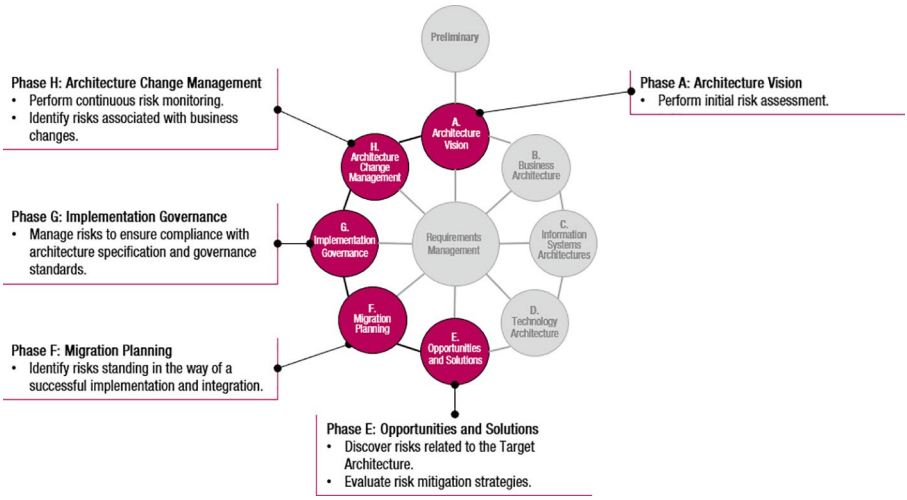


Figure 14-10. *Risk Management in specific phases of the ADM*

There is a difference between the technique called *Risk Management* and *Enterprise Risk Management*. The technique described in this chapter is a simplified version of the comprehensive Enterprise Risk Management. The latter is described in great detail in the Series Guide "Integrating Risk and Security within a TOGAF® Enterprise Architecture". A link to this document is provided in the *Further Reading* information box at the end of this section.

Risk Management in the TOGAF Standard primarily focuses on architecture project risk. This is only one type of risk. The scope of Enterprise Risk Management, as presented in the "Integrating Risk and Security within a TOGAF® Enterprise Architecture" Series Guide as part of the Enterprise Security Architecture, is much broader. It includes business, system, information, project, privacy, compliance, and organizational change risk, among other categories, too [24].

When a risk is identified in the course of architecture work, the first thing to consider is whether it is an *initial (preliminary) risk* or a *residual risk*. The former indicates a new, previously unaddressed risk, while the latter concerns a previously encountered – and possibly (partially) mitigated – risk. The approach is different for each type of risk.

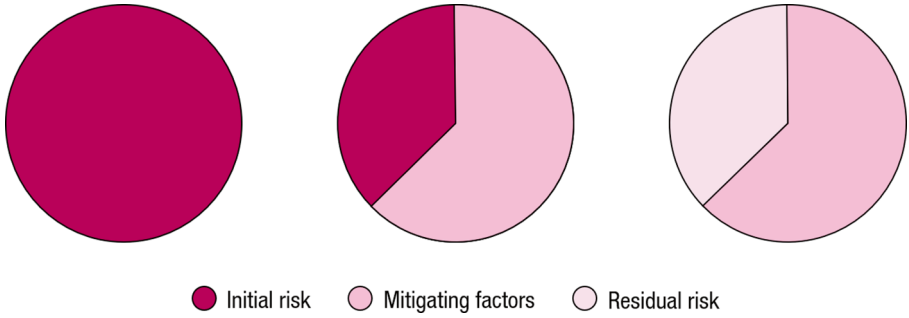


Figure 14-11. *Difference between initial and residual risks and mitigating factors*

For an initial risk – a categorization given to a risk prior to identifying and implementing mitigating actions – the Risk Management technique should be applied. Existing or residual risks have usually already been accepted and mitigating actions implemented. Figure 14-11 illustrates the difference between initial risk, residual risk, and mitigating factors.

The technique of applying Risk Management consists of a number of steps. First, risks are typically classified as time, cost, and scope, but can also include contractual, technology, and complexity risks. In addition, risks can be classified by architecture domains or segments. The Risk Management technique involves identifying the Baseline and Target Architectures and the work packages required to arrive at the Target Architecture. The consequences of *not* achieving the Target Architecture can lead to the discovery of risks.

These risks are classified by impact and frequency according to scales used in the organization. When impact and frequency are combined, they create a preliminary risk score that can be visualized in a Risk Heat Map (see Figure 14-12 in Section 14.8.6).

After the risks are identified and classified, they need to be mitigated (or at least as much of the risk as possible). This means reducing the risk to an acceptable level. Once the initial risk has been mitigated, the remaining risk is referred to as *residual risk*. The key consideration is that the mitigation effort actually reduces the impact to the organization and does not simply move the risk to another similarly high quadrant.

One final step of the Risk Management technique involves tracking and updating risk assessments over time. Once the residual risks have been accepted, the implementation of mitigating actions must be carefully monitored. This is to ensure that the organization is dealing with residual risks rather than initial risks. Phase G (Implementation Governance) is where risk monitoring takes place.



Classify the Risks: First, identify the types of risks involved. Common categories include time, cost, and scope. But don't stop there. Also consider contractual commitments, technological challenges, and the complexity of the project. In addition, consider risks specific to different parts of the architecture.

Identify the Risks: Next, review the Baseline and Target Architectures. Identify the steps required to reach the Target Architecture. By analyzing these steps, potential risks that could occur if the Target Architecture is not achieved can be identified.

Assess the Risks: Once the risks have been identified, assess how often they could occur and how much impact they could have. Use the organization's standard scales to make this assessment. Combining these factors provides a preliminary risk score that helps prioritize which risks need more attention.

Mitigate the Risks and Assess What's Left: Plan and implement actions to reduce the significant risks to acceptable levels. After these mitigation efforts, assess the remainder of the risks, known as residual risks. Ensure that the mitigation strategies effectively reduce the impact on the organization and don't just shift the risk elsewhere.

Monitor the Risks: Finally, keep track of the risks over time. Once the residual risks are accepted, closely monitor the implementation of the mitigation actions. This ongoing oversight ensures that the residual risks are managed effectively, rather than facing the initial risks all over again. This continuous monitoring happens during Phase G (Implementation Governance).

Each of the steps to Risk Management is further detailed in the sections below.



Further Reading

The Risk Management technique is further described in *The TOGAF Standard, 10th Edition*, document number C220.

More information about Enterprise Risk Management is available through the Series Guide "Integrating Risk and Security within a TOGAF® Enterprise Architecture", document number G152.

14.8.1 Risk Classification

Effective Risk Management is critical for organizations seeking to protect their objectives and operations. A structured approach to risk classification enables rapid and appropriate mitigation strategies.

A common method is to categorize risks based on their potential impact on the organization. This approach ensures that risks with significant impact are escalated to the appropriate governance levels, facilitating timely and effective responses.

Risks are often neatly categorized. This segmentation helps identify and effectively address potential pitfalls. Primary categories include

- **Time (Schedule) Risks:** These pertain to delays that can derail project timelines.
- **Cost (Budget) Risks:** These involve expenditures that exceed the allocated budget.
- **Scope Risks:** These arise when there's a deviation from the project's defined objectives.

Given the multifaceted nature of risks, organizations should also consider the following risk categories when classifying risks:

- **Contractual Risks:** These relate to potential issues arising from contractual terms and obligations. Misunderstandings or ambiguities in contracts can lead to disputes or unmet expectations.
- **Technology Risks:** As technology rapidly evolves, existing systems may become obsolete or incompatible. Keeping abreast of technological advances is critical to mitigating these risks.
- **Scope and Complexity Risks:** Projects with large or complex scope can be challenging to execute and manage. Clear definition and understanding of project boundaries is critical.
- **Environmental (Corporate) Risks:** Internal cultural or structural challenges within an organization can impede progress. Aligning organizational structures and cultures with project goals is essential.
- **People Risks:** Staffing issues, such as skills shortages or turnover, can affect project delivery. Investing in training and retention strategies helps mitigate these risks.
- **Client Acceptance Risks:** Deliverables may be met with resistance from customers or stakeholders. Engaging clients throughout the project lifecycle ensures that their needs and expectations are addressed.

Another effective strategy is to classify risks by architecture domain or segment:

- **Business Risks:** Threats that could derail business processes and objectives. Think of them as hurdles that could derail strategic goals.
- **Information Risks:** Data integrity and security concerns. It's all about making sure information is accurate and protected.
- **Application Risks:** These are potential problems within software applications. Whether it's bugs, integration problems, or performance issues, they fall into this category.
- **Technology Risks:** This domain covers vulnerabilities in the hardware and supporting technologies. It's about keeping the backbone of the IT infrastructure robust and resilient.

The Implementation Factor Catalog (see Section 12.8.2) may be supplemented by any of the classifications listed above.

Table 14-14. *Implementation Factor Catalog Supplemented by a Risk Classification*

Risk Classification	Factor Category	Factor	Description	Deduction
Environmental (corporate) risk People risk	Effect (impact)	Change in Applications	Replace HR Recruitment Application A with Application B to streamline storage of candidate information.	Need for personnel training to use new application. New application has major personnel savings and should be given priority.
Scope and complexity risk	Dependency	Consolidation of Application Services	Multiple information streams will be consolidated.	New application will automate most application services.
Business risk	Action	Introduction of New HR Recruitment Application		

The first row in the Implementation Factor Catalog (Table 14-14) is about replacing HR Recruitment Application A with Application B. The deduction of this factor shows that it is expected that there will be a *need for personnel training* to adequately handle the new application. In addition, the implementation of the new application is likely to result in a

reduction in headcount (*major personnel savings*). Because the first factor in the catalog involves *internal cultural or structural challenges* and *staffing issues that (may) affect project delivery*, it is classified as an *environmental (corporate) risk* and a *people risk*.

Organizations may have unique categorizations tailored to their specific contexts. It's advisable for the Architecture Capability of the organization to adopt or extend these classifications to maintain consistency and relevance.

14.8.2 Risk Identification

During the execution of phases B, C, and D of the Architecture Development Method, the Baseline and Target Architectures are formed. The gaps between these architecture states will undoubtedly allow one or more risks to surface. Performing Business Transformation Readiness Assessments does the same. Identifying these risks is critical because it allows organizations to develop strategies to address them throughout the transformation process.

In Phase E (Opportunities and Solutions), these risks are captured in the Implementation Factor Catalog. This is where they are further detailed. The catalog is then used during Phase F (Migration Planning) to proactively address the risks.

Capability Maturity Models (see also Chapter 15) are particularly effective for assessing specific factors related to architecture delivery. They help establish baseline and target states and outline the actions required to move to the desired state. Failure to achieve the target state may reveal potential risks that require attention.

Documenting how to approach these risks is typically done in a Risk Management Plan. Standard project management methodologies such as the Project Management Body of Knowledge (PMBOK) and PRINCE2 provide templates.

A Risk Management Plan contains at least three basic components:

- **Document Information:** Things like the project name, date, and author of the document, including the version number.
- **Approvals:** A list of the dates, names, and signatures of the people who approved the approach described in the document.
- **Approach:** A detailed description of how risks will be managed. It also includes a summary of the tools that will be used and an outline of the information that will be recorded. Perhaps more importantly, it defines the metrics that will be used to classify risks. Risk tolerance is also described in the document.

The aforementioned project management methodologies typically include procedures for contingency planning, tracking and assessing risk levels, responding to changes in risk factors, and processes for documenting, reporting, and communicating risks to stakeholders.



Further Reading

Additional information about project management methodologies can be found at these URLs:

https://www.pmi.org/standards/pmbok
https://www.axelos.com/certifications/propath/prince2-project-management

14.8.3 Initial Risk Assessment

The initial risk assessment takes place during phases E and F of the Architecture Development Method. In the Opportunities and Solutions Phase (Phase E), the Implementation Factor Catalog and the Consolidated Gaps, Solutions, and Dependencies Matrix are used to capture risks. Phase F proactively addresses these risks by creating a Risk Management Plan. This plan covers the potential risks identified during migration planning with proposed mitigation strategies.

Once risks have been identified, the first step is to determine whether they are *initial* or *residual*. The approach to these types of risks differs. Section 14.8 describes these approaches.

There are no hard and fast rules with respect to measuring impact (effect) and frequency (likelihood). The guidelines described in the TOGAF Standard are based upon existing Risk Management best practices [20].

Risks are usually classified with respect to *impact* (effect) and *frequency* (likelihood) in accordance with scales used within the organization. In most organizations, it is common to assign a value between 1 and 5 to both impact and frequency. The sample criteria shown in Tables 14-15 and 14-16 can be used to classify the impact and frequency of risks. Of course, these criteria can be adapted to the organization.

Table 14-15. *Criteria to determine Risk Impact Level*

Impact (Effect)	Impact Level	Description
Catastrophic	5	Critical financial loss that could lead to the organization's bankruptcy.
Significant	4	Causes severe financial loss in more than one line of business, resulting in lost productivity and no return on (IT) investment.
Moderate	3	Results in a small financial loss to a line of business and a reduced return on (IT) investment.
Low	2	Infers a very minor impact on the ability of a line of business to provide services and/or products, not accompanied by any type of financial loss.
Negligible	1	Assumes minimal impact on business unit's ability to deliver services and/or products.

Table 14-16. *Criteria to determine Risk Frequency Level*

Frequency (Likelihood)	Frequency Level	Description
Frequent	5	Likely to occur very often and/or continuously.
Probable	4	Occurs several times during a transformation cycle.
Occasional	3	Occurs sporadically.
Seldom	2	Possible, but unlikely to occur more than once in the course of a transformation cycle.
Unlikely	1	Not likely to occur in the course of a transformation cycle.

The two factors (impact and frequency) can be combined to provide a score and classification of the actual risk. This method is described in more detail in Section 14.8.6.1. Table 14-17 shows the possible risk classifications.

Table 14-17. *Criteria to determine Risk Classification*

Classification	Indication	Description
Critical risk	C	The transformation effort is likely to fail, with serious consequences.
High risk	H	Significant failure of parts of the transformation effort, resulting in failure to achieve certain goals.
Moderate risk	M	Noticeable failure of parts of the transformation effort, jeopardizing the success of certain goals.
Low risk	L	Certain goals will not be fully achieved.

Something very important to mention here is that the risk being measured is a risk to the organization. It is not a risk to the architecture or the work around it. The risks that emerge from the initial risk assessment are risks that the organization faces if the architecture described at this point is implemented.

When the impact and frequency values are plotted against each other as is shown in Table 14-18, it is possible to see which risk classification applies to each intersecting cell. This risk classification scheme forms the basis for the Risk Heat Map (used for plotting and visualizing risks), which is discussed in more detail in Section 14.8.6.

Table 14-18. *Risk Classification Scheme*

Impact ▼	Frequency Frequent	Probable	Occasional	Seldom	Unlikely
Catastrophic	C	C	H	M	L
Significant	C	H	M	M	L
Moderate	H	M	M	L	L
Low	M	M	L	L	L
Negligible	L	L	L	L	L

For the initial risk assessment to work, everyone in the organization must agree on the definitions of impact levels and frequency. Once this agreement is reached, the values can be used to visualize the risks encountered in a Risk Heat Map. The use of this technique is explained in more detail in Section 14.8.6.

14.8.4 Risk Mitigation and Residual Risk Assessment

Risk mitigation is a critical aspect of strategic planning for any organization. It involves identifying potential risks, developing plans to address them, and executing actions to reduce those risks to an acceptable level. Mitigation strategies can range from simply monitoring or accepting a risk to comprehensive contingency plans, such as implementing full redundancy as part of a business continuity plan. Each approach has its own scope, cost, and time implications. Given the significant impact of risk assessments, it's important to conduct them in a pragmatic yet systematic manner. Prioritizing frequent, high-impact risks ensures that each risk is addressed effectively and efficiently.

During Phase E (Opportunities and Solutions), the Implementation Factor Catalog (see Section 12.8.2) is created. This catalog is used as input to the Risk Management technique. When the architecture activities reach Phase F (Migration Planning), this is where Risk Management is incorporated into the Implementation and Migration Plan.

As described in the previous sections, a risk is first identified as an initial or preliminary risk. The classification is then extended by categorizing the risk (e.g., time, cost, scope). Next, the impact and frequency of the risk are determined. A final step in the risk identification process is to define the mitigating factors that will either eliminate the risk or reduce the risk level. If the risk is not fully mitigated, what remains of the risk is called the residual risk. It is important to re-evaluate the impact and frequency level of this residual risk to see if the mitigation effort has really made an acceptable difference. Table 14-19 shows an example of how this information can be recorded.

Table 14-19. *Risk Identification and Mitigation Assessment Matrix*

#	Risk	Preliminary Risk			Mitigation	Residual Risk		
		Imp.	Freq.	Risk Level		Imp.	Freq.	Risk Level
1	Reduction in headcount	3	4	12	Automate capabilities	2	2	4
2	Unable to use application	2	3	6	Provide training for staff	2	1	2

It is important to note whether the mitigation efforts have actually reduced the impact of the risk on the organization. The risk should not simply have been moved to a different quadrant of similar importance.

For example, if mitigating factors reduce a risk from catastrophic to critical, it is still an extremely high risk. In this case, the mitigation effort must be reconsidered. Mitigating factors should have a noticeable effect on the risks.

Part of the Risk Management technique described in this chapter is to perform a transformation risk assessment by creating a Risk Identification and Mitigation Assessment Matrix. Table 14-19 illustrates this deliverable.

The Risk Identification and Mitigation Assessment Matrix contains the *risk ID* along with a *description* of the risk. This description is based on the *Deduction column* from the Implementation Factor Catalog.

The Preliminary Risk columns contain the *impact* and *frequency values* of the initial risk. Together, they result in a risk score (*impact* x *frequency*), which is noted in the *Risk Level* column.

The Mitigation column of the Risk Identification and Mitigation Assessment Matrix contains the *mitigation factors* that will be applied to the risk, followed by the impact, frequency, and risk levels after these factors are implemented.



List All Potential Risks: Start by identifying every possible risk that could impact the architecture work. Assign a unique ID and provide a clear description for each risk.

Determine Initial Impact and Frequency: For each risk, assess

- **Impact:** How severe would the consequences be if this risk occurred?
- **Frequency:** How often is this risk likely to happen?

Multiply these two values to get an initial risk score.

Record Initial Risk Level: Based on the initial risk score, classify the risk into a category.

Plan Mitigation Strategies: For each risk, decide on actions to reduce its impact or likelihood. Document these mitigation measures.

Reassess Impact and Frequency After Mitigation: After implementing mitigation strategies, evaluate

- **New Impact:** What would be the consequences now if the risk occurred?
- **New Frequency:** How likely is the risk to happen after mitigation?

Multiply these new values to get a post-mitigation risk score.

Record Post-mitigation Risk Level: Classify the risk again based on the new score to see how the mitigation efforts have changed its severity.

14.8.5 Risk Monitoring

Residual risks must gain approval through the Governance Framework, potentially extending to corporate governance for business acceptance. Once accepted, it's crucial to meticulously monitor the execution of mitigating actions, ensuring the organization addresses residual rather than initial risks. The Risk Identification and Mitigation Assessment

Matrix serves as a governance artifact. It is maintained and updated during Phase G (Implementation Governance), where risk monitoring occurs. This phase can spotlight critical risks that remain unmitigated, possibly necessitating another full or partial Architecture Development Method cycle.

Later on, during Phase H, risk monitoring is performed with respect to *changes to the architecture*. The Architecture Change Management Phase identifies risks associated with new requirements, technologies, or business changes. It implements risk response mechanisms as part of change management.

14.8.6 Visualizing Risks

After conducting the initial risk assessment as described in Section 14.8.3, it may be a good idea to visualize the various risks identified and compare them to the organization's risk appetite or tolerance. Although the TOGAF Standard does not formally describe how risk can be applied in visualizations, there are a number of ways to approach this.

A Risk Heat Map (see Figure 14-12) is a visualization tool that helps organize, define, and quickly communicate key risks. It is an essential tool in any Risk Management toolbox. Such a heat map is often used to focus management's attention on the most important threats and opportunities and to lay the groundwork for risk responses. The heat map is based on the *risk classification scheme* (Table 14-18) that was detailed in Section 14.8.3.

Risk heat mapping is a powerful technique that adds *risk visualization* to the Risk Management technique already described in the TOGAF Standard. The Risk Heat Map shows *risk frequency* (likelihood) on the horizontal axis (x) and *risk impact* (effect) on the vertical axis (y). Together, these axes can help analyze a risk and decide what actions to take to mitigate it.

The heat map is a two-dimensional representation of data in which values are typically represented by colors (often red, green, and yellow). It allows the architect to effectively and visually communicate which risks are of most and least concern to the organization, depending on its risk appetite or tolerance. Typically, green represents low risk, yellow represents medium risk, and orange/red represents high/critical risk. This information can guide risk mitigation and minimization strategies. In the risk assessment process, visualizing risks using a heat map provides a concise, big-picture view of the entire risk landscape, opening the door for discussion as decisions are made about the likelihood and impact of risks within the organization.

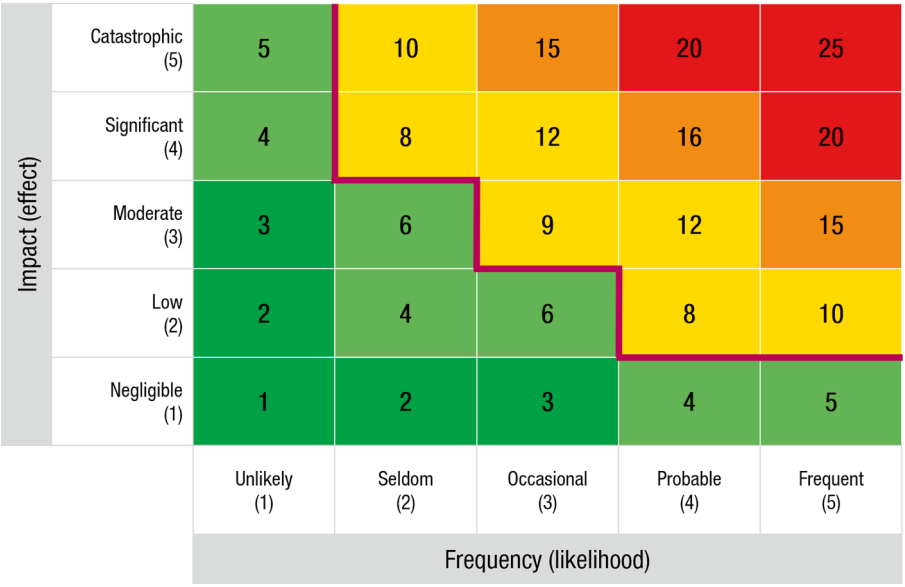


Figure 14-12. *Risk Heat Map*

The use of a Risk Heat Map, as shown in Figure 14-12, is very effective in communicating which risks have the highest significance when considering their potential impact and the likelihood of that impact.

Psychologically, people looking at the heat map are naturally drawn to the upper-right quadrant (high impact and high likelihood), while items in other quadrants receive less attention.

Calculating Risk

To use a Risk Heat Map, it's important for the organization to establish a common language for discussing risk. Section 14.8.3 outlined that terms such as *impact* or *effect* and *frequency* or *likelihood* need to be defined and used throughout the organization and in the design of the heat map. This allows everyone to be on the same page when discussing risk. Using a Risk Heat Map also requires a common understanding of the organization's *risk appetite* or *tolerance*. Risk tolerance refers to the highest level of risk that is still considered acceptable. In Figure 14-12, the risk tolerance is visualized by the maroon-colored line. All the risk values below the line are considered to be acceptable and are therefore given a green color.

Risks are generally assigned a score for risk frequency (the likelihood that the risk will occur) and the impact or effect the risk will have on the organization. Unfortunately, there is no scientific approach to assigning these scores. Common sense is the best way to go here.

The formula for calculating risk is as follows:

$$\text{Risk} = \text{Impact (effect)} \times \text{Frequency (likelihood)}$$

Once each risk has been calculated, it can be plotted on the Risk Heat Map as shown in Figure 14-13.

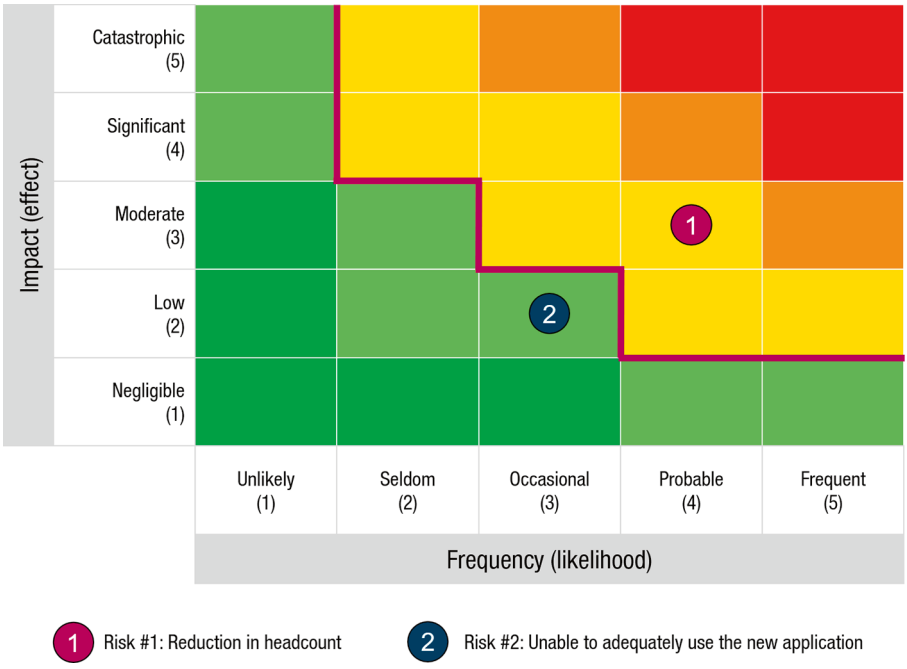


Figure 14-13. *Risk Heat Map with mapped risks*

Figure 14-13 shows the Risk Heat Map described earlier with two risks. Each risk is represented by a number in a circle corresponding to the risk. For the first risk (Risk #1), the impact is estimated to be *moderate* (3) and the likelihood of the risk occurring is estimated to be *probable* (4). Using the formula above, the risk score is 12, meaning *medium* risk classification. The second risk (Risk #2) has a *low* impact (2) and a frequency (likelihood) of *occasional* (3). Therefore, the risk score for the second risk is 6, classifying it as a *low* risk.

Scoring each risk in this way and plotting them in the heat map provides a good picture of which risks need the most attention. Implementation plans can be adjusted accordingly.

Adding Risks to Diagrams

Although the TOGAF Standard does not formally describe how risk can be applied in visualizations, the ArchiMate standard (another standard from The Open Group) does. As mentioned in Chapter 1, Section 1.3, this book is not about the ArchiMate standard, nor is it about modeling ethics. However, to give a sense of how risks can be incorporated into diagrams, the following example is given.

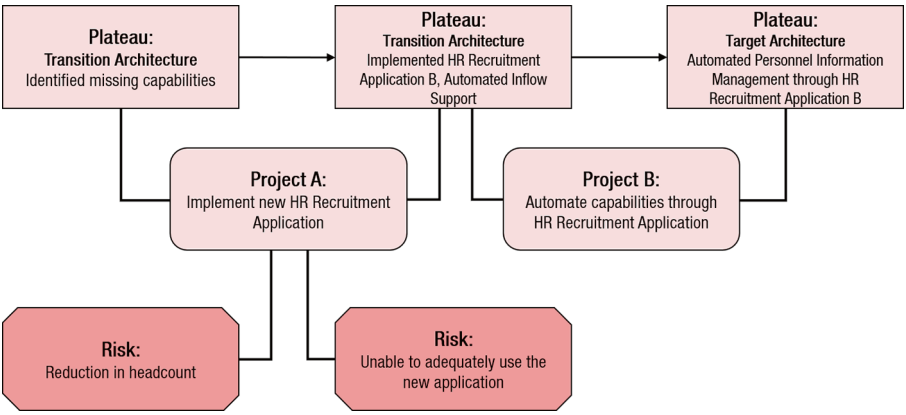


Figure 14-14. *Plateau Planning Diagram with added risks*

In Chapter 12, Section 12.8.2, a Plateau Planning Diagram was shown. This diagram illustrated the use of projects instead of gaps and work packages. Figure 14-14 shows the addition to this diagram of risks derived from the Implementation Factor Catalog regarding the implementation of HR Recruitment Application B.

There is, of course, much more to be said about modeling risk in diagrams. However, this book deals with the TOGAF Standard, not the ArchiMate standard. Therefore, the example above is for illustrative purposes only.



Further Reading

For more information about visualizing risks, refer to the URL below:

https://bizzdesign.com/blog/the-value-of-enterprise-architecture-in-managing-risk-compliance-and-security/

14.9. Architecture Alternatives and Trade-Offs

Architecture development entails more than pursuing a single solution or solution direction. In many architectural scenarios, it is even uncommon to find a single solution that addresses all stakeholders' concerns. Given the involvement of numerous stakeholders in a project or in the development and implementation of architecture, it is essential to propose not just one solution but at least two. By presenting stakeholders with multiple alternatives, architects can identify any underlying agendas or principles that could potentially influence the architecture.

The TOGAF Standard provides a structured approach to explore various alternatives and engage stakeholders in meaningful discussions. This technique is called Architecture Alternatives and Trade-Offs and is illustrated in Figure 14-15.

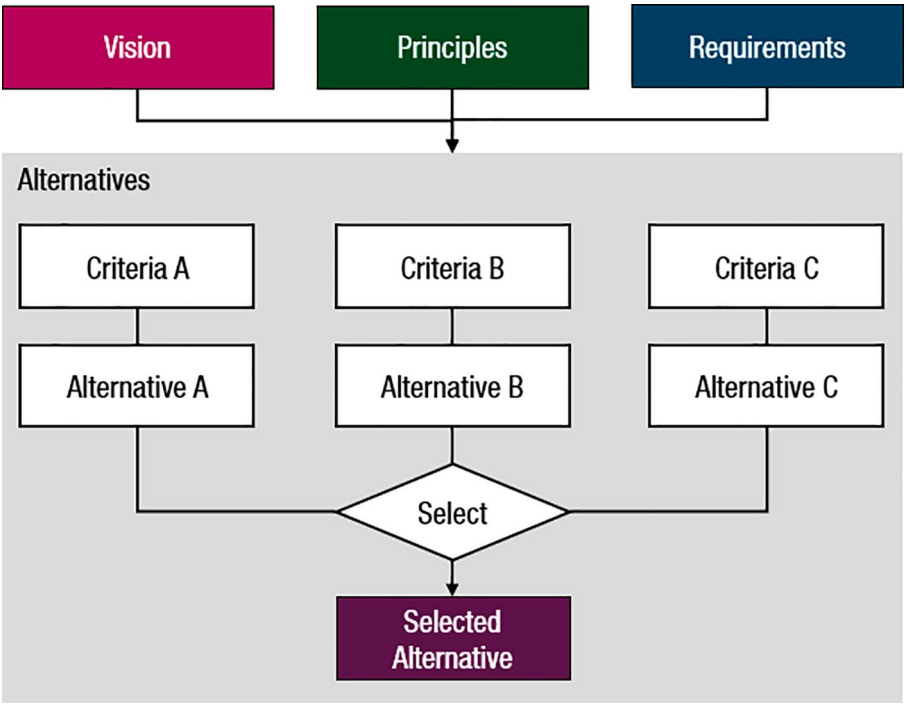


Figure 14-15. *Architecture Alternatives and Trade-Offs method*

The approach begins by leveraging the organization's vision, principles, requirements, and other pertinent information to establish a set of criteria suitable for assessing different alternatives. Then, based on these criteria, different alternatives are formulated, leading to a deeper understanding of each option. Finally, the process involves selecting the most appropriate alternative or combining elements from multiple options to develop a proposed solution. This approach involves conducting thorough analysis to support informed decision-making. This method is versatile and can be applied at various stages and levels of architectural development.



Gather Information: Begin by gathering all the important information about the organization – its vision, guiding principles, specific needs, and any other relevant details. This foundational knowledge will help to set clear criteria to evaluate different options.

Explore Different Options: With these criteria in hand, brainstorm various alternatives. This exploration will deepen the understanding of each possibility and how it aligns with the organization's goals.

Make an Informed Decision: Analyze each option thoroughly based on the established criteria. Then, choose the most suitable alternative or combine elements from multiple options to craft the best solution. This method is flexible and can be applied at different stages and levels of architecture development.

Unfortunately, there is no fixed set of criteria that can be used to determine the alternatives. Much depends on the architecture work or project being undertaken. As a general rule, it is advisable to consider such things as Architecture Principles, requirements, the architecture vision, and stakeholder concerns when developing the alternative scenarios.

Of course, each alternative has advantages and disadvantages. These need to be discussed with the stakeholders involved and agreement reached on the final chosen alternative.

Selecting the right alternative isn't just a checkbox exercise. It involves a nuanced process influenced by a tapestry of inputs. Architecture Principles, stakeholder concerns, requirements, and the overarching vision all play pivotal roles in shaping the criteria used for evaluation.

Each alternative brings its own set of advantages and challenges. Engaging stakeholders in open discussions ensures that these pros and cons are thoroughly examined and understood. Sometimes, to paint a clearer picture, it might be necessary to introduce additional viewpoints or perspectives. This helps stakeholders delve deeper into the alternatives, uncovering dependencies, assessing risks, and navigating uncertainties.

The types of alternatives to consider depend on a number of criteria:

- **Flexibility:** How adaptable is the alternative to changing business needs?
- **Time and Cost:** What are the implications in terms of implementation timeframes and associated costs, including transitional phases and stabilization periods?
- **Benefit Realization:** Over what period can the reaping of the benefits of this alternative be expected?
- **Architectural Alignment:** Does the alternative adhere to the established architecture styles or guidelines?
- **Delivery Method:** Are existing solutions being reused, developed anew, or are off-the-shelf products being purchased?
- **Business Impact:** How minimal is the disruption to existing business capabilities during implementation?
- **Risk Profile:** What risks are associated with this alternative, and what mitigation strategies are in place?

In my other book, *Getting Started with Enterprise Architecture* [9], I gave an example of a fictional organization called Lemon-A-de. This organization spent the early days of its existence developing a strategy (see Figure 14-16).

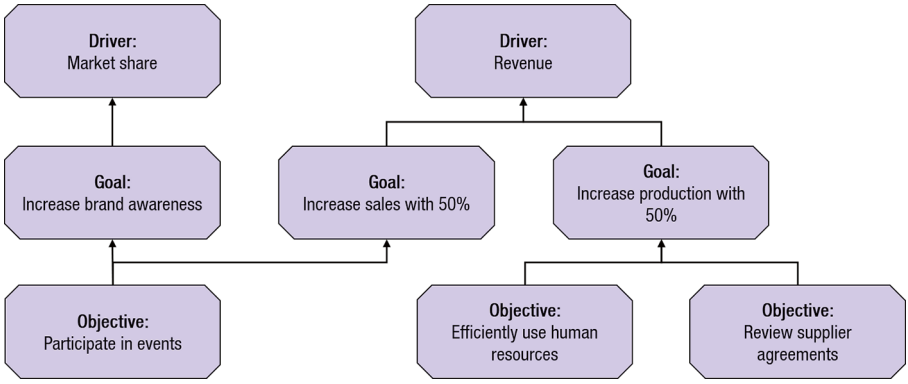


Figure 14-16. *Lemon-A-de's strategy*

The organization's architect was asked to visualize the impact of the strategy. To do this, this person looked at the goals the organization had in mind and translated them into the activities (initiatives) the company needed to develop to achieve them.

The architect's research showed that it was not immediately possible to achieve all the goals at once, given all kinds of organizational constraints. So the architect presented two scenarios to the CEO.

The Architecture Alternatives and Trade-Offs method was used here (see Figure 14-17).

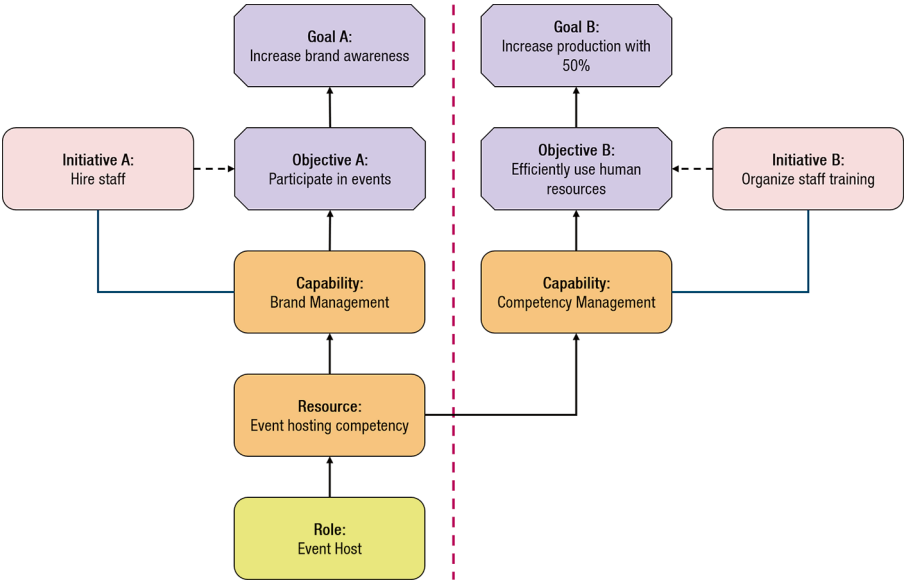


Figure 14-17. *Architecture Alternatives and Trade-Offs diagram*

Figure 14-17 shows that achieving the goal of *increasing brand awareness* (Goal A) is considered achievable if the organization commits to *hiring additional staff* (Initiative A). This will allow the organization to *plan and host several events* (Objective A), which will ultimately lead to an increase in brand awareness. An additional business role must also be defined and filled to make this possible.

The second scenario, which will achieve the goal of *increasing production by 50%* (Goal B), consists of the activity of *organizing staff training* (Initiative B).

Participation in training will allow for more efficient deployment of personnel within the organization (Objective B), which may allow for more effective work. In all likelihood, this will lead to the goal being achieved.

Both scenarios – alternatives – come at a cost. The first scenario will increase brand awareness, but has the downside that this awareness may lead to increased demand for the product. This demand may not be able to be met immediately due to staff inefficiencies.

Scenario 2, which focuses on using staff more efficiently by providing appropriate training, will lead to increased production. The downside is that if brand awareness lags, this production stock will not be sold within the desired timeframe, leaving the company with an inventory.

The Architecture Alternatives and Trade-Offs method provides insight into the advantages and disadvantages of all the scenarios or solutions mentioned and tries to satisfy all the wishes and requirements of the parties involved. Often, the application of this method leads to a phased roadmap, in which, in this example, both scenarios are eventually realized.

Presenting multiple, well-considered alternatives gives the organization confidence that scenarios have been considered from a variety of perspectives. It also demonstrates that all stakeholder concerns have been considered.



Further Reading

For more information about the Architecture Alternatives and Trade-Offs method, refer to *The TOGAF Standard, 10th Edition*, document number C220.

14.10. Summary

The Architecture Development Method is often regarded as the core of the TOGAF Standard. It consists of a cyclic method that helps organizations align business and IT needs effectively. Various techniques are applied throughout the Architecture Development Method cycle to ensure efficient decision-making, risk mitigation, and stakeholder engagement. These techniques support different phases of the Architecture Development Method and enhance architectural practice.

When the TOGAF Standard talks about techniques, it does not always mean a *specific way of doing things*. What is really meant by techniques is an elaboration of a topic or component that is frequently used within the framework or during the implementation of the Architecture Development Method. A recurring topic that requires further explanation and clarification in order to be properly applied.

Some topics are essential aspects of developing an architecture. However, they are not necessarily a specific technique. Stakeholder Management and Risk Management both are essential, but they are not techniques. However, the TOGAF Standard still refers to them as if they are.

The framework distinguishes between eight techniques:

- **Architecture Principles:** Fundamental rules and guidelines that influence architecture development and governance.
- **Stakeholder Management:** A method for identifying, analyzing, and managing stakeholders.
- **Architecture Patterns:** Provide standardized solutions to common architectural problems.
- **Gap Analysis:** Identifies differences between the Baseline and Target Architectures.

- **Interoperability Requirements:** Determine interoperability requirements.
- **Business Transformation Readiness Assessment:** Describes a technique for identifying business transformation issues.
- **Risk Management:** Ensures that potential issues are identified and mitigated before impacting the architecture.
- **Architecture Alternatives and Trade-Offs:** Describes a technique to identify alternative Target Architectures and perform trade-offs between the alternatives.

Principles play a crucial role in guiding decision-making and ensuring alignment between business objectives and strategies. Principles are often established within different domains and at different levels of an organization.

The TOGAF Standard distinguishes between Enterprise Principles and Architecture Principles:

- **Enterprise Principles:** High-level, general guidelines that shape the overall strategy, culture, and decision-making framework of an organization.
- **Architecture Principles:** More specific guidelines that direct the design and development of Enterprise Architecture.

Within an Enterprise Architecture approach, principles serve as a foundation for designing flexible and scalable architectures that align with the aforementioned business goals and objectives. They ensure compliance with regulations and industry standards and are able to reduce complexity and duplication by enforcing standardization and interoperability.

Stakeholder Management is the process of identifying and understanding the needs, interests, and expectations of an organization's various stakeholders and planning and implementing actions to address those interests and improve cooperation and communication with them. The stakeholder concept is closely related to governance in Enterprise Architecture. Understanding who the stakeholders are and what their interests are helps architects establish governance mechanisms to address their concerns, maintain transparency, and ensure accountability.

The TOGAF Standard recommends incorporating stakeholder analysis during Phase A (Architecture Vision) of the Architecture Development Method and updating it throughout each following phase. This form of continuous engagement ensures that new stakeholders are identified and their concerns are addressed as the project progresses through phases such as Phase E (Opportunities and Solutions), Phase F (Migration Planning), and Phase H (Architecture Change Management).

The "Architecture Patterns" section is a different story. Although this topic is included as such in the TOGAF Standard, it states that this technique has not yet been incorporated into the Architecture Development Method [20]. In fact, the TOGAF Standard is indicating that it is not useful enough. Why it is included in the framework remains a mystery. It is possible that a future version of the TOGAF Standard will provide more information on this subject.

A *gap analysis* is a process that identifies the difference between an organization's Baseline and Target Architectures in terms of its capabilities, processes, applications, and technology. The purpose of a gap analysis is to identify areas where improvements are needed to achieve the desired future state. The results of the gap analysis are used to inform subsequent phases of the Architecture Development Method, such as phases B, C, and D, where the detailed architecture is developed based on the architectural vision.

During Phase E (Opportunities and Solutions) and Phase F (Migration Planning) of the Architecture Development Method, five artifacts can be created to help fine-tune the migration planning:

- **Phase E (Opportunities and Solutions):**
  - **Implementation Factor Catalog:** This catalog documents factors impacting the architecture Implementation and Migration Plan.
  - **Consolidated Gaps, Solutions, and Dependencies Matrix:** Can be used to group the gaps identified in the gap analysis results and assess potential solutions and dependencies to one or more gaps.
  - **Architecture Definition Increments Table:** Used to plan a series of Transition Architectures outlining the status of the Enterprise Architecture at specified times.
- **Phase F (Migration Planning):**
  - **Transition Architecture State Evolution Table:** This table shows the proposed state of the architectures at various levels using the defined taxonomy.
  - **Business Value and Risk Matrix:** Effective assessment of business value can be achieved by creating a matrix based on a value index dimension and a risk index dimension.

The idea behind the *interoperability technique* is to compare a set of entities and determine what form and degree of interoperability is required between them.

One major advantage of applying the interoperability technique is enhanced integration and communication between systems. Interoperability ensures that different entities (such as people, systems, applications, and services) can effectively exchange and use information, reducing data silos and integration complexity.

In the early phases of the Architecture Development Method, particularly Phase A (Architecture Vision) and the subsequent architecture definition phases (phases B–D), organizations conceptualize and design innovative solutions that often involve significant changes. These changes are not limited to technological shifts as they have a profound impact on the workforce, culture, and existing skills and competencies within the organization.

To ensure successful implementation of the architecture transformation in phases E and F, organizations must conduct a comprehensive *Business Transformation Readiness Assessment*. This is a collaborative effort between the organization's staff, particularly human resources, lines of business, and (IT) planners. A Business Transformation Readiness Assessment table typically includes various factors that determine an organization's preparedness for a significant change.

Risk Management is an integral part of architecture development. As such, it is addressed in several phases (Phase A and phases E–H) of the Architecture Development Method. Applying this technique ensures that risks are identified, assessed, and mitigated as part of the architecture development process.

There are two types of risks:

- **Initial (Preliminary) Risk:** A new, previously unaddressed risk.
- **Residual Risk:** A previously encountered – and possibly (partially) mitigated – risk.

For an *initial risk* – a categorization given to a risk prior to identifying and implementing mitigating actions – the Risk Management technique should be applied. Existing or *residual risks* have usually already been accepted and mitigating actions implemented.

A common method is to categorize risks based on their potential impact on the organization. This approach ensures that risks with significant impact are escalated to the appropriate governance levels, facilitating timely and effective responses.

The initial risk assessment takes place during phases E and F of the Architecture Development Method. Risks are usually classified with respect to *impact* (effect) and *frequency* (likelihood) in accordance with scales used within the organization. The two factors (impact and frequency) can be combined to provide a score and classification of the actual risk.

Risk mitigation is a critical aspect of strategic planning for any organization. It involves identifying potential risks, developing plans to address them, and executing actions to reduce those risks to an acceptable level. If the risk is not fully mitigated, what remains of the risk is called the residual risk. It is important to re-evaluate the impact and frequency level of this residual risk to see if the mitigation effort has really made an acceptable difference.

After conducting the initial risk assessment, risks can be visualized and compared to the organization's risk appetite or tolerance. Although the TOGAF Standard does not formally describe how risk can be applied in visualizations, there are a number of ways to approach this. For example, creating a Risk Heat Map allows the visualization of risks.

Architecture development entails more than pursuing a single solution or solution direction. In many architectural scenarios, it is even uncommon to find a single solution that addresses all stakeholders' concerns. The TOGAF Standard provides a structured approach to explore various alternatives and engage stakeholders in meaningful discussions. This technique is called *Architecture Alternatives and Trade-Offs*. The approach leverages the organization's vision, principles, requirements, and other pertinent information to establish a set of criteria suitable for assessing different alternatives. Based on these criteria, different alternatives are formulated. The process involves selecting the most appropriate alternative or combining elements from multiple options to develop a proposed solution.